26. Performance Evaluation of Different Garbage Collection Algorithm Based On Programming Languages Drawback/ Trade-off

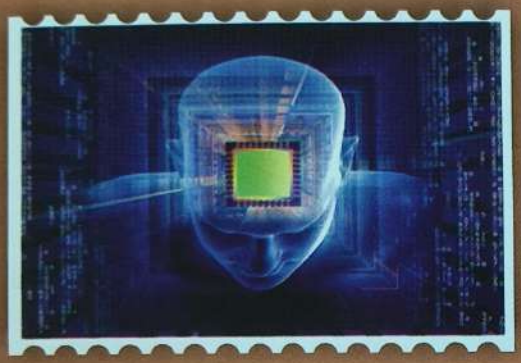
Avanaksh Singh Sambyal, Jitendra K. Chauhan and Prikhshayat Singh*
Institution of Electronics and Telecommunication Engineers,
2 Institutional Area, Lodhi Road Delhi 110003, India, Centre for
Development of Advance Computing (CDAC), Noida,
Uttar Pradesh
avasisam@gmail.com

ABSTRACT

Garbage collection is a problem related with dynamic memory management. It is the memory required from heap by the programmer to do a certain task. This concept was first utilized in early 60s that is in Lisp programming language and then before 90s, there was concept of object oriented approach used in C++ and after that in Java too. Here, in memory compaction techniques was utilized. Compaction deals with used and unused space required by certain program/s. Suppose in array implementation using heap memory, we have used some space from heap declaring variable as pointer but other space are unknown to us. If we are unable to release this memory or pages by fault, then this leads to the problem known as garbage collection. If there should be some means that the space collected is in garbage or not. Here comes the picture of algorithms. The Algorithm used in this field are many but for the performance evaluation measure, the algorithm such as mark and sweep and baker's semi-space and reference counting will be used. The complexity measure of these algorithms such as best case, worst case, average case, etc. was also studied in this paper.

Key words: Garbage collection, mark and sweep algorithm.

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